Geometric series

1 Find the sum of the first 12 terms of the series: $5 + 10 + 20 \ldots$

2 Find the sum of the first 5 terms of the geometric sequence defined by the following. Round your answers to two decimal places.

a $a = 2.187$ and $r = 1.134$

b $a = -4.186$ and $r = -2.848$

3 Find the sum of the first 11 terms of the series: $1 - 2 + 4 \ldots$

4 Find the sum of the first 7 terms of the geometric series: $64 + 16 + 4 \ldots$

5 Find the sum of the first 5 terms of the series: $24 - 12 + 6 \ldots$
Round your answer to the nearest whole number.

6 Find the sum of the first 7 terms of the series: $8 + 1 + \frac{1}{8} \ldots$
Round your answer to three decimal places.

7 Consider the series: $5 + \frac{5}{2} + \frac{5}{4} \ldots$

a Find the common ratio, r.

b Find the sum of the first 10 terms, rounding your answer to one decimal place.

8 The first 3 terms of a sequence are $x,\ \frac{2}{5}x^2,\ \frac{4}{25}x^3$.

a Write a simplified expression for the sum of the first n terms of this sequence.

b Evaluate the sum when $x = 5$ and $n = 9$.

9 Consider the series: $\frac{1}{4} - \frac{1}{5} + \frac{1}{16} - \frac{1}{25} + \frac{1}{64} - \frac{1}{125} + \ldots$
Form an expression for the sum of the first $2n$ terms of the series.

10 For each of the following series:

i Find n, the number of terms in the series.

ii Find the sum of the series.

a $4 - 8 + 16 - \ldots - 2048$

b $64 + 16 + 4 + \ldots + \frac{1}{}$

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11 Find n, the number of terms, in each of the following series:

a The sum of n terms in the geometric series $2 + 10 + 50 + \ldots$ is $195\,312$.

b The sum of n terms in the geometric series $5, -20, 80, \ldots$ is $262\,145$.

c The sum of n terms in the geometric series $16 + 4 + 1 + \ldots$ is $\frac{1365}{64}$.

d The sum of n terms in the geometric series $40, -8, \frac{8}{5}, \ldots$ is $\frac{20\,832}{625}$.

12 The sum of the first 6 terms of a geometric series is 28 times the sum of its first 3 terms. Find r, the common ratio.

13 The sum of the first 9 terms of a geometric series is $436\,905$, and the common ratio is $\frac{1}{4}$. Find a, the first term in the series.

14 The first term of a geometric sequence is 9 and the fourth term is 243. What is the sum of the first 13 terms of the corresponding series?

Infinite series

15 What condition must be satisfied by an infinite geometric series in order for its sum to exist?

16 For each of the following infinite geometric sequences:

i Find the common ratio.

ii Determine whether the sum is divergent or convergent.

a $11, 22, 44, 88, \ldots$

b $3, -12, 48, -192, \ldots$

c $-40, -20, -10, -5, \ldots$

d $256, 64, 16, 4, \ldots$

17 For each of the following infinite geometric sequences:

i Find the common ratio, r.

ii Find the limiting sum of the series.

a $2, \frac{1}{2}, \frac{1}{8}, \frac{1}{32}, \ldots$

b $125, 25, 5, 1, \ldots$

c $16, -8, 4, -2, \ldots$

d $28, -7, \frac{7}{4}, -\frac{7}{16}, \ldots$

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18 For a particular geometric sequence, $a = 2$ and $S_\infty = 4$.

a Find r, the common ratio.

b Find the first 3 terms in the sequence.

19 Consider the infinite sequence: $1, x-7, (x-7)^2, (x-7)^3, \ldots$
Determine the range of values of x if the infinite series has a limiting sum.

20 The limiting sum of the infinite sequence $1, 4x, 16x^2, \ldots$ is 5. Find the value of x.

21 Consider a geometric series in which all terms are positive that has first term a and common ratio r. The sum of the infinite series is 768. The sum of the first 4 terms is 765.
Find the first three terms of the sequence.

22 Consider the infinite geometric series: $6 + 2 + \frac{2}{3} + \frac{2}{9} + \ldots$

a Find the number of terms, n, that would be required to give a sum of $\frac{177\,146}{19\,683}$.

b Find the infinite sum of the series.

23 Consider the infinite geometric series: $5 + \sqrt{5} + 1 + \ldots$

a Find the common ratio, r, expressing your answer with a rational denominator.

b Find the limiting sum, expressing your answer with a rational denominator.

24 Find the limiting sum of the infinite series $\frac{1}{5} + \frac{3}{5^2} + \frac{1}{5^3} + \frac{3}{5^4} + \frac{1}{5^5} + \ldots$

Sigma notation

25 Evaluate:

a $\sum_{i=1}^{4} 4^i$

b $\sum_{j=1}^{5} 243\left(\frac{5}{3}\right)^j$

c $\sum_{j=1}^{5} 32\left(\frac{1}{2}\right)^j$

d $\sum_{i=3}^{8} (-3)^i$

e $\sum_{i=2}^{7} \left(-2^i\right)$

f $\sum_{i=1}^{5} \left(2\,(3)^{i-1}\right)$

g $\sum_{j=4}^{8} \left(\frac{1}{4}\,(3)^{j-1}\right)$

h $\sum_{j=1}^{4} \left(-3\left(\frac{1}{3}\right)^j\right)$

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26 Write the series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \ldots + \frac{1}{1024}$ using sigma notation.

27 Find the value of the following:

a $\sum_{i=1}^{\infty} 3\left(\frac{1}{4}\right)^{i-1}$ **b** $\sum_{i=1}^{\infty} 5\left(-\frac{1}{4}\right)^{i-1}$ **c** $\sum_{i=1}^{\infty} \frac{1}{8}\left(-\frac{1}{2}\right)^{i-1}$ **d** $\sum_{i=1}^{\infty} -\frac{1}{2}\left(\frac{5}{7}\right)^{i-1}$

e $\sum_{i=1}^{\infty} (0.9)^{i}$ **f** $\sum_{k=1}^{\infty} 9^{-k}$

Repeating decimals

28 The repeating decimal $0.8888\ldots$ can be expressed as a fraction when viewed as an infinite geometric series.

a Using fractions, write the first five terms of the geometric sequence representing $0.8888\ldots$

b State the values of the first term a and the common ratio r of this sequence.

c Calculate the infinite sum of the sequence as a fraction.

29 The repeating decimal $0.444444\ldots$ can be expressed as a fraction when viewed as an infinite geometric series.

a Express the first two decimal places, 0.44, as an unsimplified fraction.

b Express the second two decimal places, 0.0044, as an unsimplified fraction.

c Express the third two decimal places, 0.000044, as an unsimplified fraction.

d Now, state the values of the first term a and the common ratio r of the sequence formed from these first three terms.

e Calculate the infinite sum of the sequence as a fraction.

30 For each of the following repeating decimals:

i Rewrite the decimal as a geometric series.

ii Express the decimal as a fraction.

a $0.6666\ldots$ **b** $0.06666\ldots$

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31 Consider the sum $0.25 + 0.0025 + 0.000025 + \ldots$

a Express the sum as a decimal.

b Now, express the sum as a fraction.

32 Consider the repeating decimal $0.2222\ldots$ in the form

$$\frac{2}{10} + \frac{2}{100} + \frac{2}{1000} + \frac{2}{10\,000} + \ldots$$

Rewrite the repeating decimal as a fraction.

33 Consider the repeating decimal $0.575\,75\ldots$ in the form

$$0.57 + 0.0057 + \ldots$$

Rewrite the repeating decimal as a fraction.

34 Find the value of $0.994 + 0.000\,994 + 0.000\,000\,994\ldots$

Applications

35 At the start of 2014, Pauline deposits \$5000 into an investment account. At the end of each quarter, she makes an extra deposit of \$700.
The table below shows the first few quarters of 2014. All values in the table are in dollars:

Quarter	Opening Balance	Interest	Deposit	Closing Balance
Jan - Mar	5000	200	700	5900
Apr - Jun	5900	236.00	700	6836.00
Jul - Sep	6836.00	273.44	700	7809.44

a Find the quarterly interest rate.

b Write an expression for the amount in the account at the end of the first quarter.

c Write an expression for the amount in the account at the end of the second quarter.

d Write a similar expression for the amount in the account at the end of the third quarter.

e Write an expression for the amount in the investment account after n quarters.

f Now, determine the total amount in Pauline's account at the beginning of 2016 to the nearest dollar.

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36 Iain opens an account to help save for a house. He opens the account at the beginning of 2017 with an initial deposit of $\$50\,000$ that is compounded annually at a rate of 5.9% per annum. He makes further deposits of $\$3000$ at the end of each year.

a Write an expression for the amount in the account after:

i 1 year **ii** 2 years **iii** 3 years

b The amount in the account after n years can be expressed as:

$$50\,000 \times (1.059)^n + 3000 \times (1.059)^{n-1} + \ldots + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$$

This can be written as $50\,000 \times (1.059)^n$, plus the sum of a geometric sequence. Write an expression representing this sum.

c Now, determine the total value of Iain's savings in 2024, to the nearest dollar.

37 Lisa invests $\$30\,000$ at a rate of 1.5% per month compounded monthly. Each month, she withdraws $\$600$ from her investment after the interest is paid and the balance is reinvested in the account.

a Write an expression for A_1, the balance of the account after 1 month.

b Write an expression for:

i A_2 **ii** A_3

c Write an expression for A_n.

d Calculate the amount Lisa has saved after 3 years.

38 To save up to buy a car, Laura opens a savings account that earns 6% per annum compounded monthly.
She initially deposits $\$1400$ when she opens the account at the beginning of the month, and then deposits $\$165$ at the end of every month.

a The amount in the account after n months can be expressed as the nth term of a geometric sequence plus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months.

b Now, determine the amount Laura has saved after 3 years.

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39 Rochelle invests $\$190\,000$ at a rate of 7% per annum compounded annually, and wants to work out how much she can withdraw each year to ensure the investment lasts 20 years.

a Write an expression for A_1, the balance of the account after 1 year. Use x to represent the amount to be withdrawn each year.

b Write an expression for:

i A_2 **ii** A_3

c Write an expression for A_n.

d Now, determine Rochelle's annual withdrawal amount.

40 Mario invests $\$110\,000$ at a rate of 16% per annum compounded quarterly and wants the investment to last 25 years.

a The amount in the account after n quarters can be expressed as the nth term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n quarters. Use x to represent the amount to be withdrawn each quarter.

b Now, determine Mario's quarterly withdrawal amount.

41 Emma invests $\$190\,000$ at a rate of 3% per annum compounded monthly, and wants to work out how much she can withdraw each month to ensure the investment lasts 30 years.

a The amount in the account after n months can be expressed as the nth term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months. Use x to represent the amount to be withdrawn each month.

b Now, determine Emma's monthly withdrawal amount.

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